RBE/CS 549 · WORCESTER POLYTECHNIC INSTITUTE

Deep Visual Inertial Odometry

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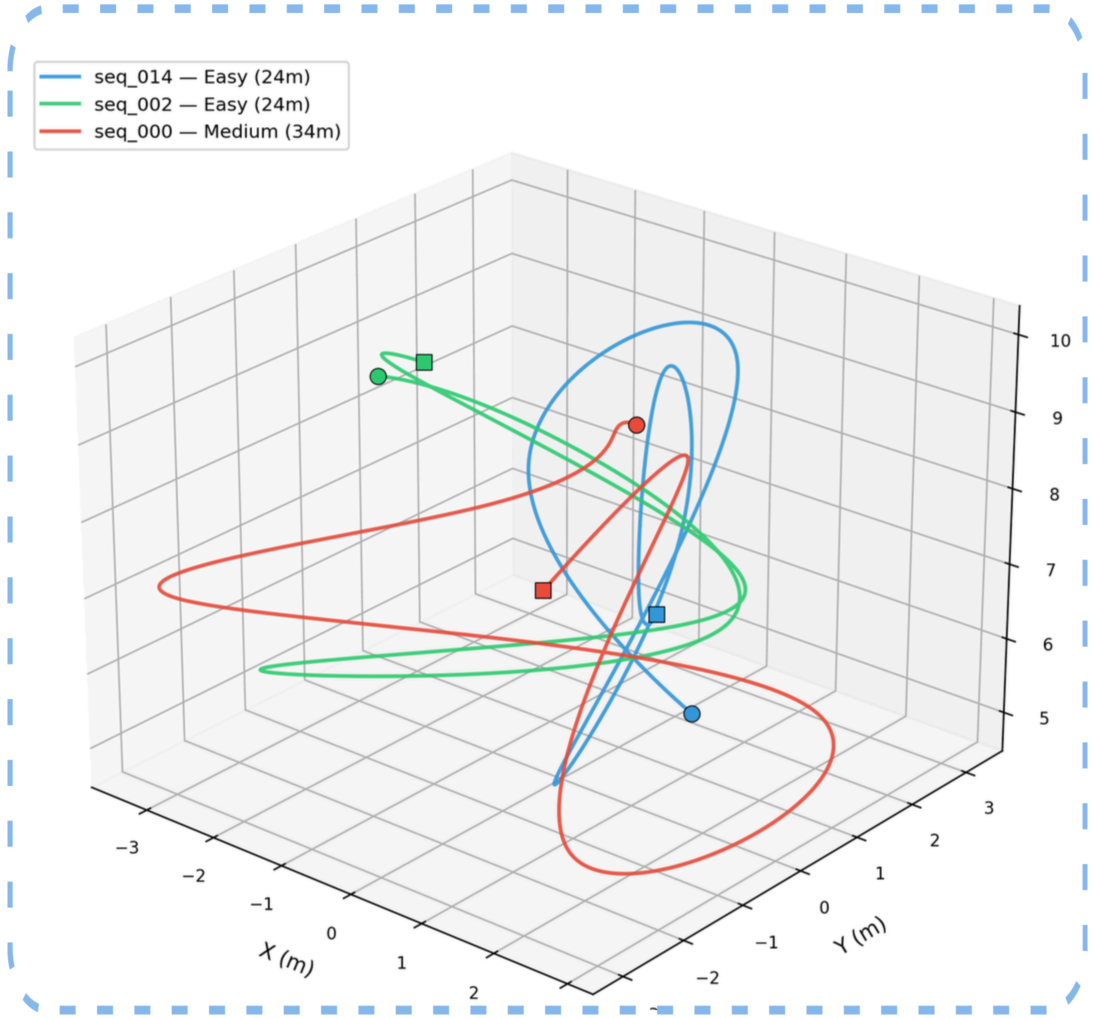
Robotics Engineering, WPI

Synthetic VIO dataset

Example texture

Example texture

Sample trajectory



Scene

- Blender
- single textured plane
- 35 textures (21 OSM + 14 procedural)

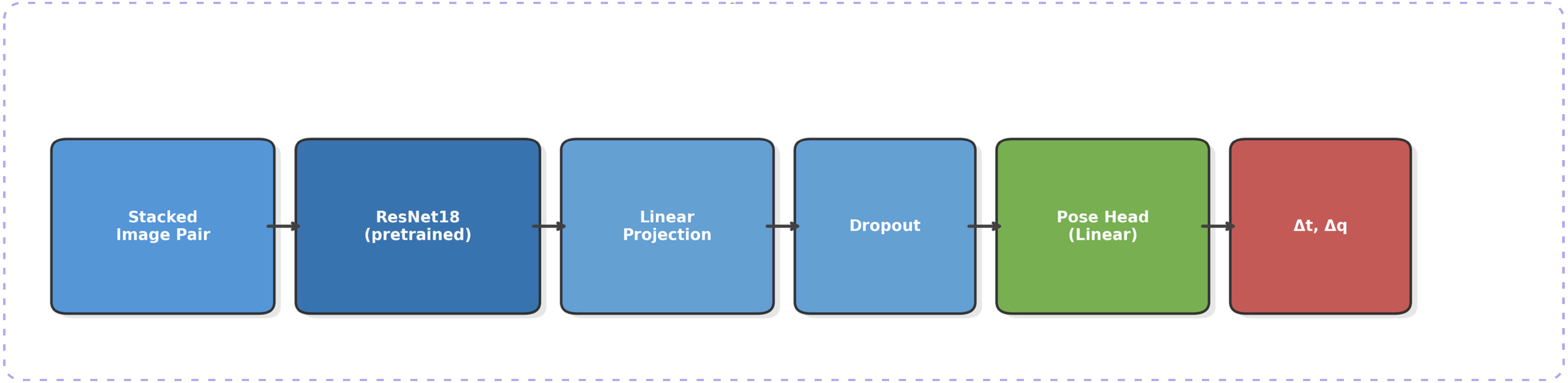
Sensors

- Camera 100 Hz; 320×240
- IMU 1000 Hz
- Gauss-Markov bias drift

Trajectories

- 60 train / 10 test
- cubic spline
- randomized pose, texture, scale

Vision-only architecture

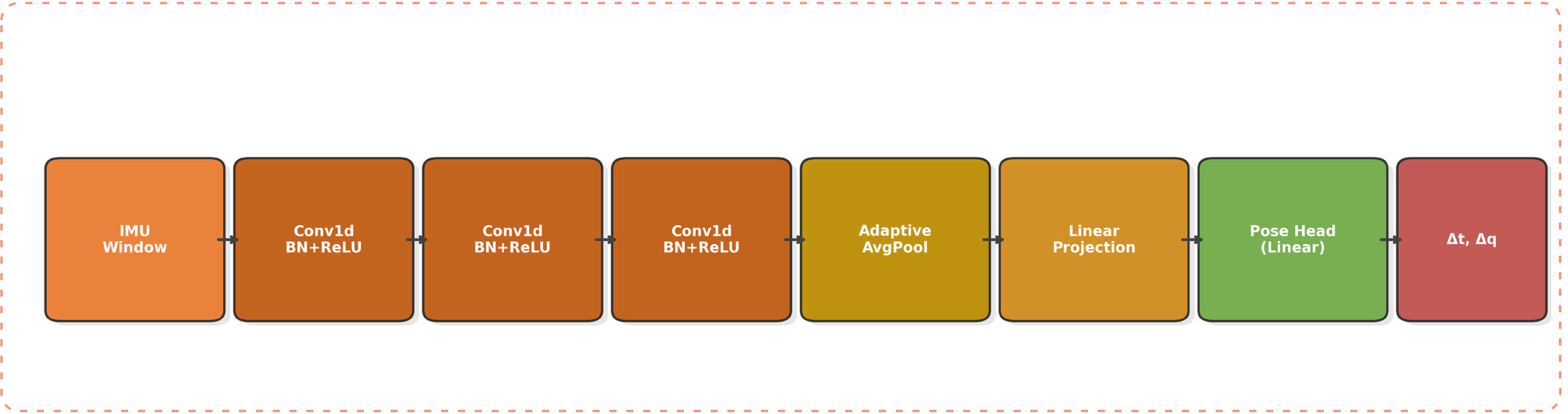


- Pretrained ResNet18, conv1 weights replicated 3→6 channels and halved to preserve activation scale
- Shared pose head (Linear→7, q-normalized) across all three networks. ~11M params.

$$\mathcal{L} = \mathbf{MSE}(\Delta\mathbf{t}) + \lambda \cdot \min(\|\mathbf{q}_{\text{pred}} - \mathbf{q}_{\text{gt}}\|^2, \|\mathbf{q}_{\text{pred}} + \mathbf{q}_{\text{gt}}\|^2)$$

$$\lambda = 10$$

Inertial-only architecture

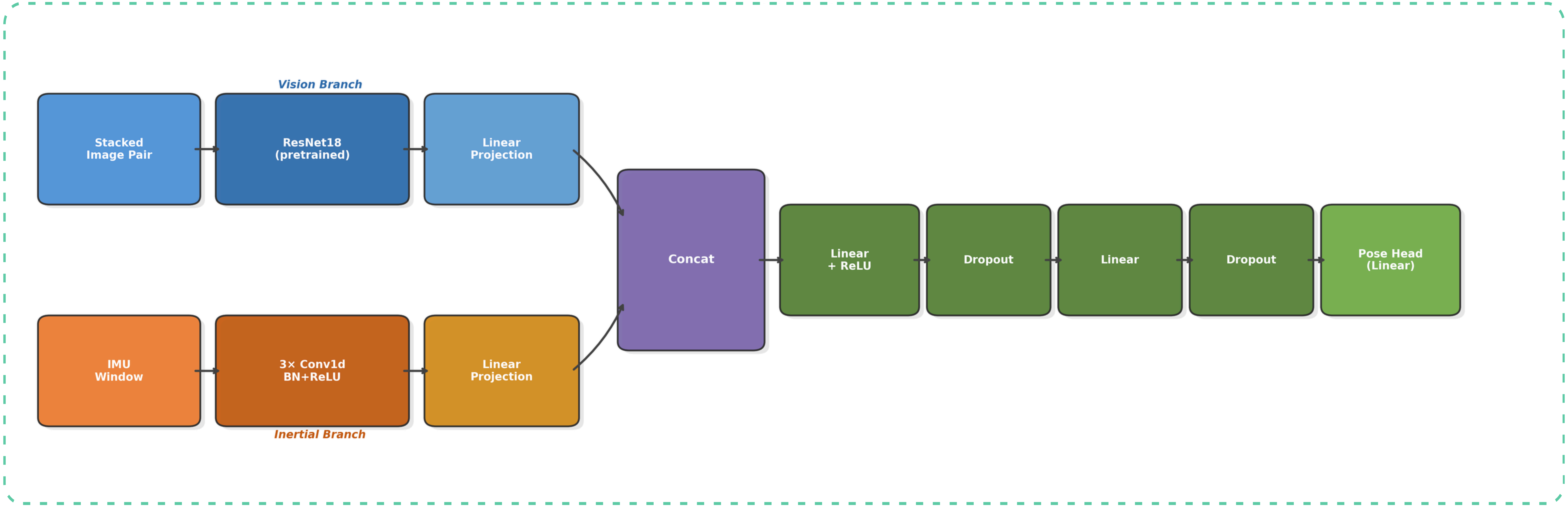


- Temporal 1D conv over 10-sample IMU windows.
- Padding preserves sequence length — architecture is agnostic to window size. ~48K params.
- BN handles the gyro/accel scale mismatch

$$\mathcal{L} = \mathbf{MSE}(\Delta\mathbf{t}) + \lambda \cdot \min(\|\mathbf{q}_{\mathbf{pred}} - \mathbf{q}_{\mathbf{gt}}\|^2, \|\mathbf{q}_{\mathbf{pred}} + \mathbf{q}_{\mathbf{gt}}\|^2)$$

$$\lambda = 10$$

Visual-inertial fusion architecture

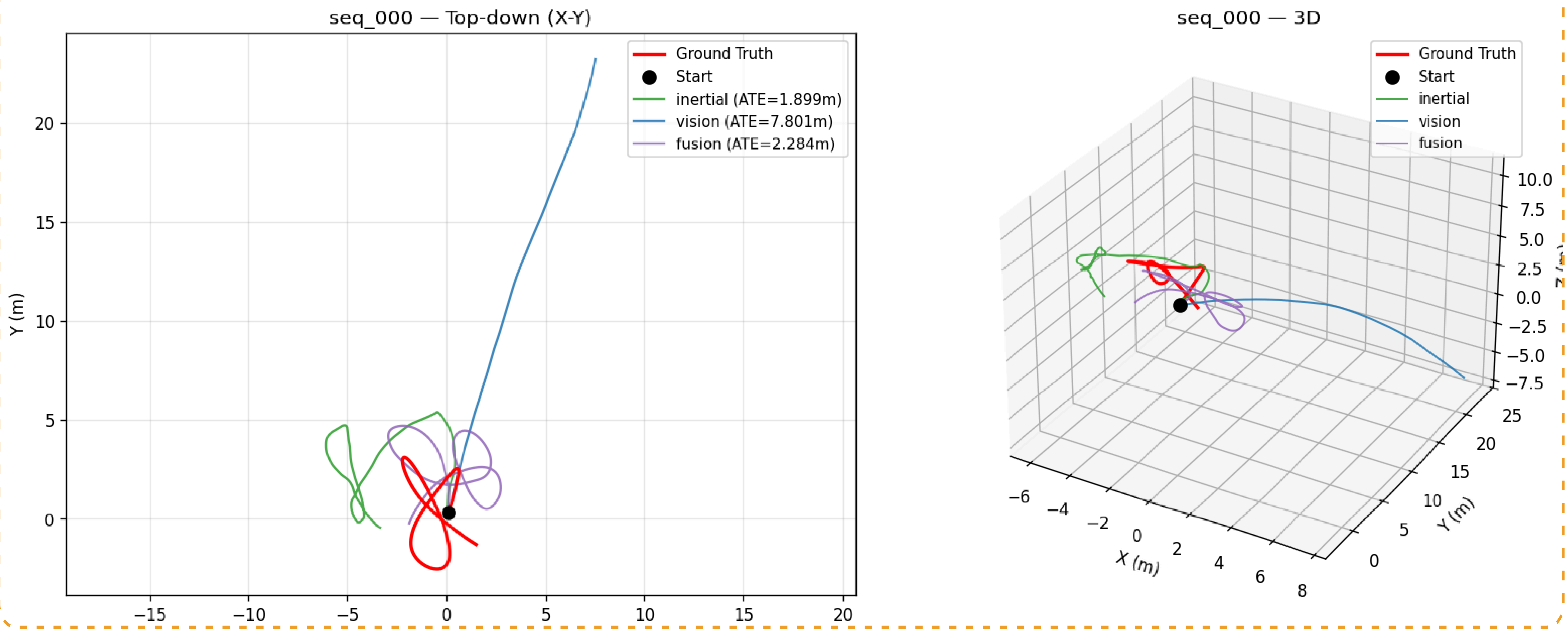


- Late fusion via concatenation. Dropout 0.3 — V1 fusion overfitted at epoch 1.
- ~11.2M params

$$\mathcal{L} = \mathbf{MSE}(\Delta\mathbf{t}) + \lambda \cdot \min(\|\mathbf{q}_{\text{pred}} - \mathbf{q}_{\text{gt}}\|^2, \|\mathbf{q}_{\text{pred}} + \mathbf{q}_{\text{gt}}\|^2)$$

$$\lambda = 10$$

Results — V1



1.89m

Inertial

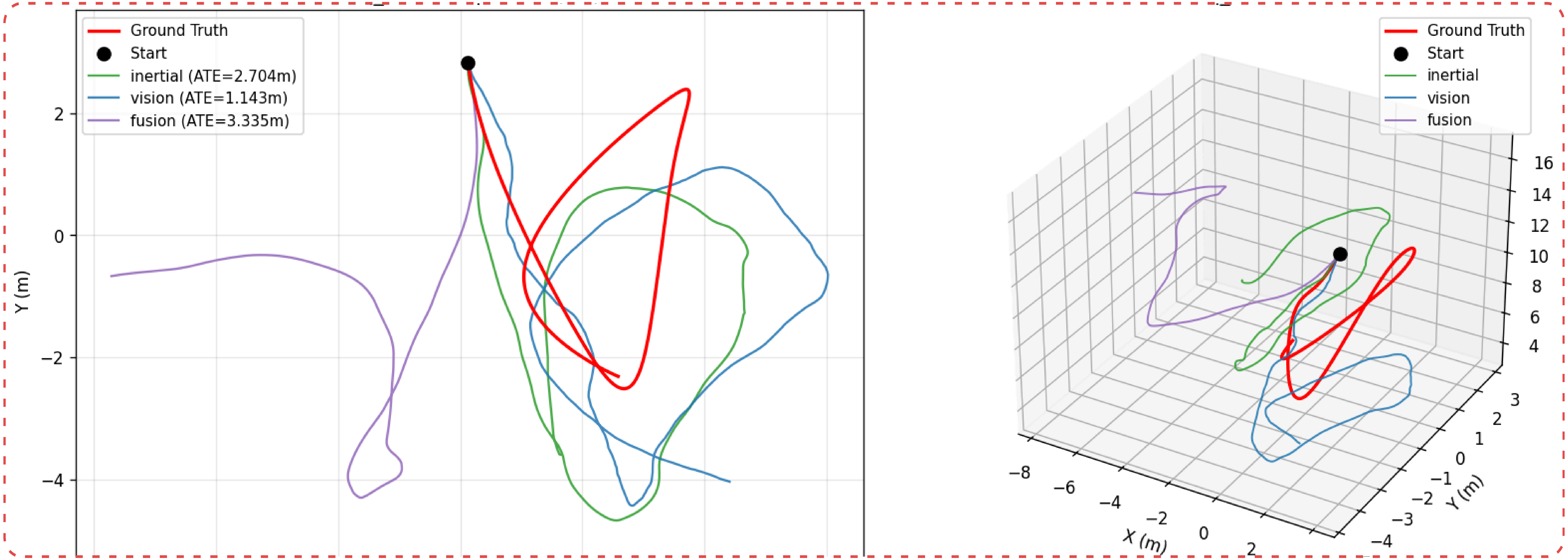
6.56m

Vision

2.58m

Fusion

Results — V2



1.95m

Inertial

1.82m

Vision

2.45m

Fusion

Things we observed

Reference Paper : DeepVIO Liming Hanet.al 2019

CNN is **optimal** due its inherent nature of better local context with less data.

Used ResNet18 as a Vision Backbone

V1's vision encoder had **Mode collapse** that was rectified by ResNet’s pretrained setup.

Doubled Dataset (30 → 60 scenes)

To support ResNet18's 11M parameters and prevent overfitting.

Why did we skip LSTM used in DeepVio - No Memory

Given just two frames (frame_t and frame_{t-1}), what's the relative motion between them - Needs No memory

Future work

Hyperparameter tuning

Better fusion strategy

Homography approach